

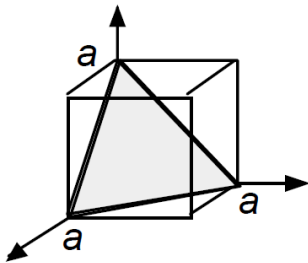
EE204 Introduction to Semiconductor Devices

HW 1

1.

Calculate the surface density of atoms on the shaded plane of the unit cell if the crystal structure is

- a) simple cubic,
- b) body-centered cubic,
- c) face-centered cubic.



$$[1/(\sqrt{3}a^2), 1/(\sqrt{3}a^2), 4/(\sqrt{3}a^2)]$$

2.

For the unit cell of the silicon crystal with lattice constant of $5.43 \text{ \AA}$,

- a) determine the number of atoms in the unit cell,
- b) calculate the shortest distance between any two atoms,
- c) calculate the volume density of silicon atoms (number of atoms/cm³) in the crystal,
- d) calculate the mass density of silicon, given that the atomic weight of silicon is 28.09 and Avogadro's number is 6.02×10^{23} atoms or molecules/mole
- e) calculate the density of valence electrons in silicon.

$$[8, 2.35 \text{ \AA}, 5.0 \times 10^{22} \text{ atoms/cm}^3, 2.33 \text{ g/cm}^3, 2.0 \times 10^{23} \text{ cm}^{-3}]$$

3.

The lattice constant of GaAs is $5.65 \text{ \AA}$,

- a) determine the number of Ga atoms and As atoms per cm³
- b) calculate the mass density of GaAs crystal, given that the atomic weights of Ga and As are 69.7 and 74.9 respectively
- c) calculate the density of valence electron in GaAs

$$[2.22 \times 10^{22} \text{ atoms/cm}^3, 5.33 \text{ g/cm}^3, 1.78 \times 10^{23} \text{ electrons/cm}^3]$$

4.

Gallium arsenide (GaAs) has an energy bandgap of 1.42 eV at room temperature. For light incident on the semiconductor, what is the maximum wavelength to create electron-hole pairs?

[875 nm]

5.

Draw the energy band diagram of a semiconductor with and without the application of an electric field ξ . Show that the electric field ξ is related to the electron energy E by

$$\xi = \frac{1}{q} \frac{dE}{dx}$$

[Optional]

6.

Considering the $E - k$ diagram in Fig 3.16 (L3 – Energy Band) explain the direct and indirect bandgap semiconductors. Explain also, why indirect bandgap semiconductor is not suitable for light emitting devices.